POREMAPS: Open-source permeability solver for pore-scale resolved, anisotropic, evolving porous media

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Motivation

POREMAPS is a free and open-source Finite Difference solver for computing Stokes flow in various complex porous materials, primarily designed for large micro X-Ray Computed Tomography (µXRCT) datasets. The MPI parallelized software fast and precisely delivers intrinsic permeability tensors of e.g. reservoir rocks in a wide porosity range and is suitable for investigations regarding anisotropy as well as simulation campaigns for materials with evolving pore space.

Governing Equations

Nondimensional, incompressible Stokes and continuity equations:

$$\mathbf{0} = \mu \operatorname{div}(\operatorname{grad} \mathbf{v}) - \operatorname{grad} p$$

Continuity equation: $\operatorname{div} \mathbf{v} = 0$

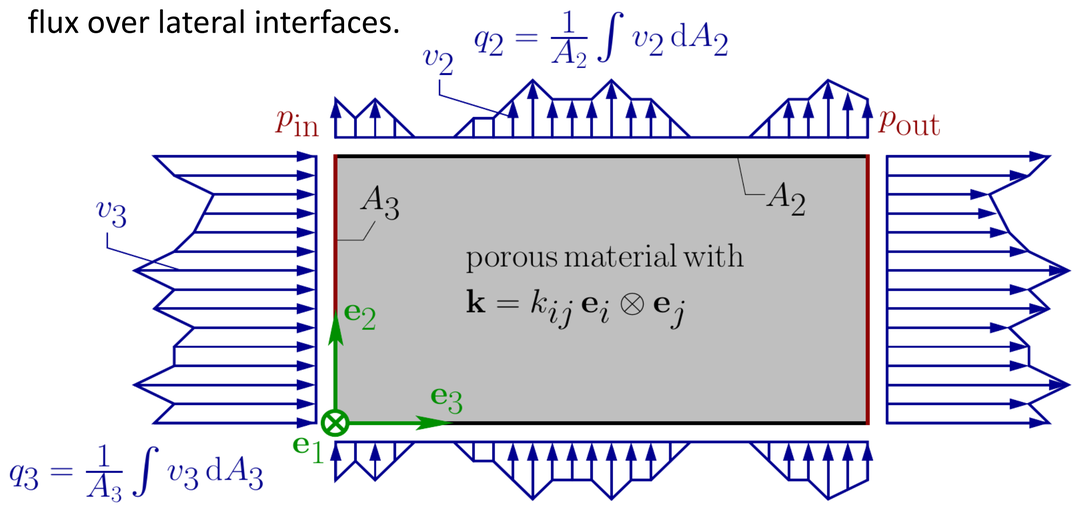
Creeping flow regime: $\mathrm{Re} = \frac{\rho_0 \mathcal{V} \mathcal{L}}{\mu} < 1$

Permeability Computation

Three numerical experiments with preset pressure gradients in all spatial directions allow for the computation of the full intrinsic permeability tensor

$$\mathbf{k} = k_{ij}\, \mathbf{e}_i \otimes \mathbf{e}_j = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix} \mathbf{e}_i \otimes \mathbf{e}_j$$

and principal permeabilities including principal axes by measuring the flow rates over the domain boundaries. Periodic boundary conditions allow for flux over lateral interfaces.



Intrinsic permeability is computed as follows:

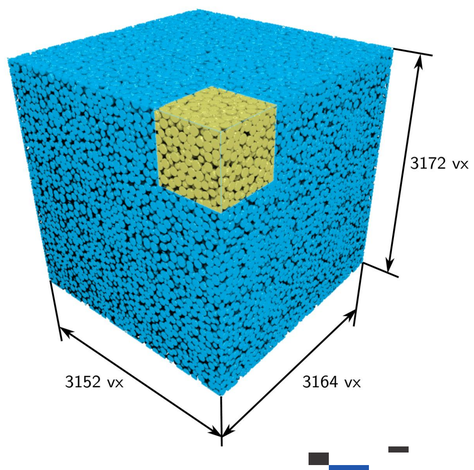
$$q_i \mathbf{e}_i = \frac{1}{\mu} k_{ij} (\mathbf{e}_i \otimes \mathbf{e}_j) \cdot h_k \mathbf{e}_k = \frac{1}{\mu} k_{ik} h_k \mathbf{e}_i$$

$$\text{with} \quad h_i = -\frac{\partial p}{\partial x_i}$$

Porous Reservoir Rock Benchmarks

Sample	ϕ	Voxel size	Domain size
Sphere packing	0.34	7.0µm	$788 \times 791 \times 793$
Berea	0.18	2.114µm	$1024 \times 1024 \times 1024$
Fontainebleau	0.09	2.072µm	$1024 \times 1024 \times 1024$

Sample	k_{11} from [2]	k_{11} (POREMAPS)
Sphere packing	$2.438 - 2.903 \cdot 10^{-10}\,\mathrm{m}^2$	$2.512 \cdot 10^{-10}\,\mathrm{m}^2$
Berea	$4.569 - 6.889 \cdot 10^{-13}\,\mathrm{m}^2$	$5.772 \cdot 10^{-13}\,\mathrm{m}^2$
Fontainebleau	$0.642 - 1.411 \cdot 10^{-13}\,\mathrm{m}^2$	$9.200 \cdot 10^{-14}\,\mathrm{m}^2$



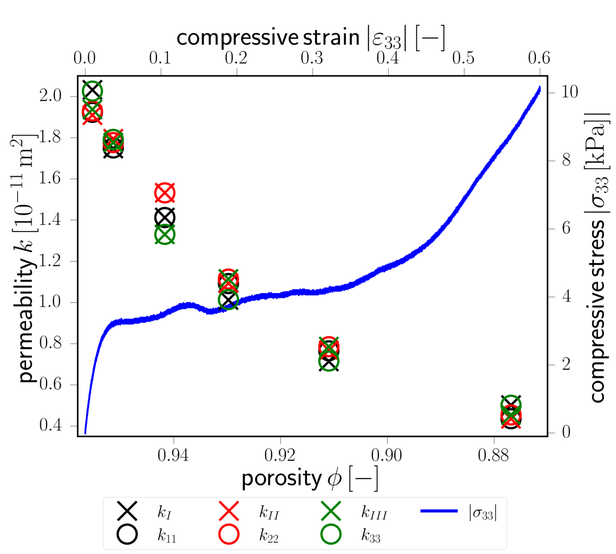
Large µXRCT datasets with $> 3000^3$ voxels

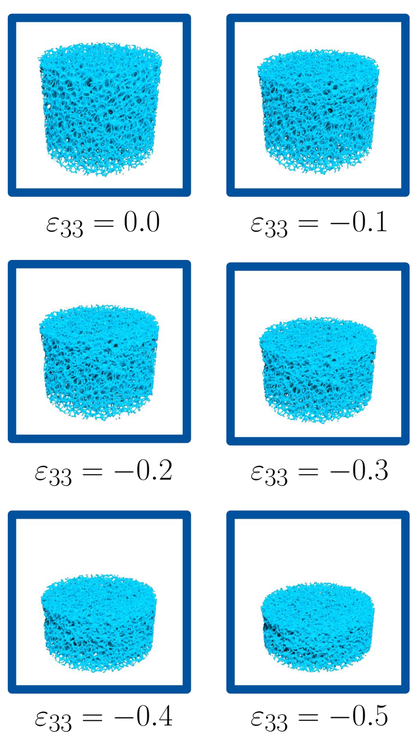
Technical Aspects

- Finite Difference discretization on MAC grid
- Open-source and free software project written in C++
- MPI parallelized for multi-node/distributed memory systems
- Operates at the pore scale using binarized 8-bit voxel-based data directly obtained from µXRCT scanners
- Scalability demonstrated in [1]

Permeabilty Evolution During Uniaxial Loading

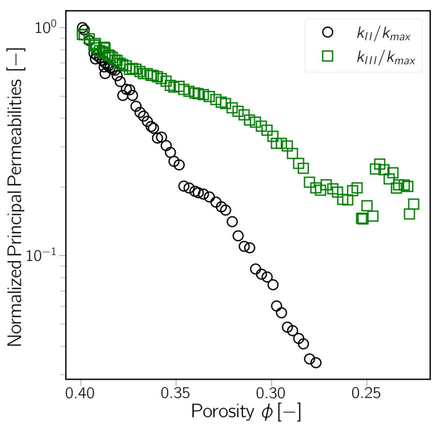
- Open-cell PUR foam sample
- Snapshots at different compression loads
- Contrary to initial expectation, no developing anisotropy observed

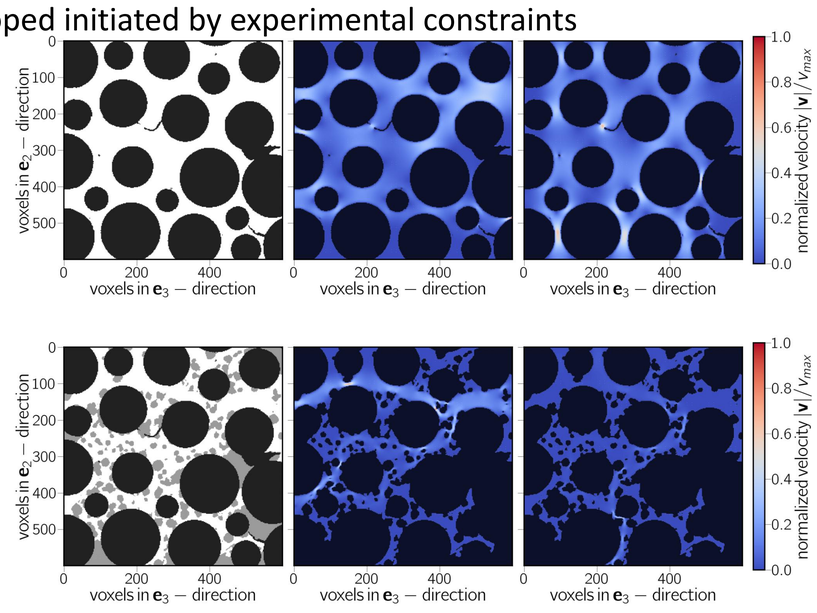




Permeability Altering During Calcite Precipitation

- Investigate enzymatically induced calcite precipitation and coherent clogging of porous material
- 128 simulations from single snapshots during experiment
- Strong anisotropy developed initiated by experimental constraints





Code Repository

Research Paper

Data, Code & Protocol Availability

Data and Code are available via the Data Repository of the University of Stuttgart (DaRUS) DOI: 10.18419/darus-3676 .

References

[1] D. Krach, M. Ruf, and H. Steeb, InterPore J, 2(1), IPJ260225–7, 2025.
[2] Saxena, N., et al., Adv. Water Resour. 109, 211-235, 2017.

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